

An arithmetic lower-envelope comparison for central quotient volumes

Full jets, finite determinant bounds, and an exact quadratic benchmark

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This calculation constructs upper bounds for the original central volume $\log(V_q/V_{2q-1})$. Its input is the proved lower envelope and Laplace estimate of the complete *Arithmetic Endpoint Bounds* source, equations (29)–(32). The comparison acts on the original polynomial and quotient maps, including all multiplicities and phases. The strongest displayed bound retains an explicit positive matrix: its arithmetic moments stop at degree $2q$, while its comparison moments continue to $4q-2$. A second bound uses only the supplied envelope constants and two Laplace values. The explicit scalar estimate has quadratic, rather than logarithmic, growth per quotient direction. An exact auxiliary calculation determines that scale and supplies a monic family relating its full jets to the original quartet jets. No bound contradicting the quartet lower estimate is asserted.

1 Original source and every jet coordinate

Fix a complete quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$, and common complete zero order $m \geq 1$ of $g = 2\xi$. Keep its monic polynomial h , its entire quotient $v_h = g/h$, and

$$w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad k \geq 3. \quad (\text{AU.1})$$

Write $c = k/2$ and keep the original observation $S = c + iu$. The source form is $\int \overline{P(c + iu)} Q(c + iu) m_k(u) du$, conjugate-linear in its first argument. Its constant norm is μ_h^k . The coefficient map and its inverse are

$$\Psi : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad P \longmapsto P(c + iu), \quad f \longmapsto f((S - c)/i). \quad (\text{AU.2})$$

These are inverse ring homomorphisms by substitution. Multiplication by S is carried to $c + iM_u$, and $\partial_u^d \Psi(P) = i^d \Psi(P^{(d)})$. On the ordered degree- N coefficient spaces the matrix is triangular with diagonal $1, i, \dots, i^N$, so its determinant has modulus one. The original quotient volumes therefore have exactly the same values in the transported remainder basis; no mass or norm is divided out.

Let $\ell = 1 + k(m-1)$ and use the ordered distinct grid

$$\lambda_{ab} = c + (2a-k)\delta + i(2b-k)\gamma, \quad \zeta_{ab} = (\lambda_{ab} - c)/i = (2b-k)\gamma - i(2a-k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{AU.3})$$

The original cyclic relation and its transported monic polynomial are

$$q = \ell(k+1)^2, \quad \chi(S) = \prod_{a,b} (S - \lambda_{ab})^\ell, \quad \psi(u) = i^{-q} \chi(c + iu) = \prod_{a,b} (u - \zeta_{ab})^\ell. \quad (\text{AU.4})$$

The grid is invariant under complex conjugation in the u chart, so ψ is real monic. Equivalently the original coefficient identity is $\bar{\chi}(k-S) = (-1)^q \chi(S)$. In particular the original norm multiplier is $\bar{\chi}(c-iu)\chi(c+iu) = \psi(u)^2$, with the source multiplication still carrying the unit i^q .

For clarity the preceding cyclic degree also follows from the original tensor algebra. In a tensor of k length- m local factors, the sum of the k nilpotent generators has power $k(m-1)$ nonzero: its coefficient on the product of their top powers is $(k(m-1))!/((m-1)!)^k$, nonzero in $\mathbb{C}$. Its next power is zero by total degree. Every grid sum in (AU.3) occurs, since the two signs in each quartet factor can be chosen independently. Direct products of local factors give the same maximal length ℓ at every grid sum. Distinct pairs have distinct real and imaginary parts. The minimal polynomial of the original sum is therefore exactly (AU.4), with no lost multiplicity.

Put $\mathcal{E} = \mathbb{C}[u]/(\psi)$, with ordered remainder coefficients $1, u, \dots, u^{q-1}$. For $N \geq q-1$ let J_N be remainder and B_N multiplication by ψ in the ordered monomials. Monic division gives the exact sequence

$$0 \longrightarrow \mathbb{C}[u]_{\leq N-q} \xrightarrow{B_N} \mathbb{C}[u]_{\leq N} \xrightarrow{J_N} \mathcal{E} \longrightarrow 0. \quad (\text{AU.5})$$

The source on the left is zero if $N = q-1$. Precisely, the coefficient isomorphisms are $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}[u]_{\leq N}$, $a \mapsto \sum_{j=0}^N a_j u^j$, and $\kappa : \mathbb{C}^q \rightarrow \mathcal{E}$, $x \mapsto [\sum_{j < q} x_j u^j]$. The polynomial quotient map is $\kappa J_N \iota_N^{-1}$ and the polynomial multiplication map is $\iota_N B_N \iota_N^{-1}$. When a matrix formula is evaluated on a polynomial below, these displayed coefficient isomorphisms are the specified identifications. The original S chart uses the same coefficient isomorphisms with u^j replaced by S^j . The complete raw derivative map is

$$\mathbf{E}_N f = (f^{(d)}(\zeta_{ab}))_{a,b,0 \leq d < \ell}, \quad (\mathbf{E}_N)_{(a,b,d),j} = \begin{cases} j! \zeta_{ab}^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d. \end{cases} \quad (\text{AU.6})$$

Taylor expansion at each distinct centre shows that its kernel is $(\psi) \cap \mathbb{C}[u]_{\leq N}$. Let $\mathbf{Z} = \mathbf{E}_{q-1}$. Then $\mathbf{E}_N = \mathbf{Z} J_N$ and $\mathbf{Z}$ is invertible. Indeed a degree below q polynomial with all those zero raw jets is divisible by degree- q ψ and is zero. For the fixed row order,

$$\det \mathbf{Z} = \left(\prod_{a,b} \prod_{d=0}^{\ell-1} d! \right) \prod_{(a,b) < (a',b')} (\zeta_{a'b'} - \zeta_{ab})^{\ell^2}. \quad (\text{AU.7})$$

One proof first regards all roots as distinct variables in the ordinary Vandermonde determinant, groups each cluster, and replaces its rows by successive divided differences. Each row operation divides by exactly the corresponding within-cluster difference. Sending the nodes in a cluster to its centre replaces its divided-difference row of order d by $f^{(d)}/d!$, by the finite Taylor expansion of each monomial. Multiplying those rows by $d!$ proves (AU.7). This is an identity of polynomials after removal of the indicated factors; no source norm or zero multiplicity is approximated. In the original S coordinates the same raw jet of order d has its literal factor i^d from (AU.2).

For any positive source matrix H_N , the least quotient form is

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{AU.8})$$

Surjectivity proves $K_N > 0$. Multiplication proves $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Thus every lift $R_N x + B_N y$ has norm $x^* G_N x + y^* B_N^* H_N B_N y$, proving the minimum property. In raw jets the kernel is $\mathbf{K}_N = \mathbf{E}_N H_N^{-1} \mathbf{E}_N^*$, and $\det \mathbf{K}_N = |\det \mathbf{Z}|^2 / V_N$. Hence

$$C_k := \log(V_q/V_{2q-1}) = \log \det \mathbf{K}_{2q-1} - \log \det \mathbf{K}_q. \quad (\text{AU.9})$$

All factorials in (AU.7) cancel only in this specified ratio.

2 The lower envelope as an explicit positive comparison form

The supplied arithmetic proof establishes $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, $B = 42 + 2 \deg h = 42 + 8m$, and

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad \alpha = \pi/2. \quad (\text{AU.10})$$

Set $a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}$ and $d_k = k - 2 > 0$. Define the comparison measure, retaining its literal mass, by

$$d\nu_k(u) = a_k e^{-\alpha|u|} (d_k + |u|)^{-B} du. \quad (\text{AU.11})$$

The identity map $L^2(m_k du) \rightarrow L^2(\nu_k)$ is well defined and a contraction, since (AU.10) gives the squared-norm inequality for every measurable function. It restricts to the identity on each polynomial space and commutes with multiplication by $c + iu$, remainder (AU.5), and every raw jet (AU.6). The comparison is therefore a map on the same quotient, not a declaration that the measures coincide.

Every moment of ν_k is finite, and the following integral evaluates it with positive integrands:

$$\nu_{k,j} := \int_{\mathbb{R}} u^j d\nu_k(u) = \begin{cases} \frac{2a_k j!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-d_k t}}{(\alpha + t)^{j+1}} dt, & j \text{ even,} \\ 0, & j \text{ odd.} \end{cases} \quad (\text{AU.12})$$

To prove it, repeated integration by parts gives $\int_0^\infty t^{B-1} e^{-y t} dt = (B-1)! y^{-B}$ for $y > 0$. Insert $y = d_k + u$ on the positive half-line. Tonelli applies because each integrand is nonnegative there. Integrating $u^j e^{-(\alpha+t)u}$ gives $j!(\alpha+t)^{-j-1}$ by the same integration by parts. Evenness supplies the factor two or the vanishing odd moment. The bound $\nu_{k,j} \leq 2a_k d_k^{-B} j! \alpha^{-j-1}$ for even j also proves finiteness directly. In particular $\nu_{k,0} > 0$ is retained, and generally differs from μ_h^k .

Let $H_N^\nu = (\nu_{k,i+j})_{0 \leq i,j \leq N}$. Its quadratic form is the ν_k integral of the polynomial's squared modulus. The weight is strictly positive everywhere, so a nonzero polynomial has positive norm. Thus $0 < H_N^\nu \preceq H_N$. Minimizing on the identical affine fibres in (AU.5) proves

$$0 < G_N^\nu \preceq G_N, \quad K_N \preceq K_N^\nu, \quad K_N^\nu = E_N (H_N^\nu)^{-1} E_N^*. \quad (\text{AU.13})$$

The inverse inequality follows also by conjugating H_N by $(H_N^\nu)^{-1/2}$ and inverting its positive eigenvalues.

Theorem 2.1 (Actual low-degree data control the higher quotient volume). *For the original quartet source and $k \geq 3$, the finite upper bound is*

$$0 \leq C_k \leq \log \det K_{2q-1}^\nu - \log \det K_q. \quad (\text{AU.14})$$

Its actual arithmetic inputs are the moments through degree $2q$ and the stated envelope constants. The comparator moments through $4q - 2$ are the explicit positive integrals (AU.12).

Proof. The first inequality follows by minimizing the same quotient norm in successively larger polynomial spaces. For the second use (AU.13) at $N = 2q - 1$ and (AU.9). Determinant is increasing on positive Hermitian forms: conjugate by the square root of the smaller one and multiply eigenvalues at least one. H_q uses precisely the original moments $0, \dots, 2q$, and all the matrices in the second term of (AU.14) have already specified finite algebraic maps from them. No higher actual moment enters. Equivalently, if $\mathfrak{D}_j^\nu = \det H_{j-1}^\nu$ and $\mathfrak{B}_j^\nu$ is the Gram determinant of the original ψ times $1, u, \dots, u^{j-1}$, the quotient-first coefficient matrix $[1, u, \dots, u^{q-1}, \psi, u\psi, \dots]$ is triangular with diagonal one. Orthogonal elimination of its relation block gives

$$V_{2q-1}^\nu = \mathfrak{D}_{2q}^\nu / \mathfrak{B}_q^\nu, \quad C_k \leq \log V_q + \log \mathfrak{B}_q^\nu - \log \mathfrak{D}_{2q}^\nu. \quad (\text{AU.15})$$

This proves the same upper bound in the exact source/relation determinants. Both the source mass and relation multiplication factors remain in their respective determinants. $\square$

The existing Gamma coefficient map also makes the cutoff in (AU.14) explicit. With $\lambda > 0$, $\alpha_\lambda = 2\lambda$, $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$ and $C_\lambda = c_\lambda^k$, define the original monic Gamma polynomials by

$$\sum_{j \geq 0} b_j^{(\lambda)}(u) z^j / j! = (1 + z^2)^{-\lambda} e^{u \arctan z}, \quad c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha_\lambda)_j}.$$

Multiplying the k generating functions and taking finite derivatives at zero gives

$$d_l = \sum_{j_1 + \dots + j_k = l} \prod_a (\alpha_\lambda)_{j_a} c_{h,j_a}, \quad \int b_l^{(k\lambda)} m_k = C_\lambda l! d_l. \quad (\text{AU.16})$$

All finite terms are integrable by the supplied exponential moments. Expand $u^{i+j} = \sum_{l \leq i+j} L_{ij,l} b_l^{(k\lambda)}(u)$ by monic subtraction. Then $(H_q)_{ij} = C_\lambda \sum_l l! L_{ij,l} d_l$. Thus the actual coefficient list $c_{h,0}, \dots, c_{h,2q}$ suffices for (AU.14). At zero degree it gives $C_\lambda c_{h,0}^k = \mu_h^k$, retaining the source mass.

3 An upper determinant depending only on the two Laplace values

Fix $0 < b < \pi/2$ and set $M_k = M_h(b)^k + M_h(-b)^k$, with the original $M_h(\pm b) = \int e^{\pm bu} w_h(u) du$. Positivity and convolution give $\int e^{b|u|} m_k(u) du \leq M_k$. The exponential series therefore gives $\int |u|^{2j} m_k(u) du \leq M_k (2j)! b^{-2j}$. For arbitrary $\tau_0, \dots, \tau_q > 0$ define

$$D_q(\tau) = M_k \left(\sum_{j=0}^q \tau_j \right) \text{diag} \left(\frac{(2j)!}{b^{2j} \tau_j} \right)_{j=0}^q. \quad (\text{AU.17})$$

For $f = \sum a_j u^j$, Cauchy–Schwarz applied to $\sqrt{\tau_j}$ and $a_j u^j / \sqrt{\tau_j}$ proves

$$|f(u)|^2 \leq \left(\sum \tau_j \right) \sum |a_j|^2 |u|^{2j} / \tau_j.$$

Integration proves $H_q \preceq D_q(\tau)$. Inverting and applying the same full jet map proves

$$\mathbf{K}_q \succeq \frac{1}{M_k \sum \tau_j} \mathbf{A}_q(\tau), \quad \mathbf{A}_q(\tau) = \mathbf{E}_q \text{diag} \left(\frac{\tau_j b^{2j}}{(2j)!} \right) \mathbf{E}_q^* > 0. \quad (\text{AU.18})$$

Consequently an upper bound without any unevaluated actual polynomial moment is

$$C_k \leq q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{A}_q(\tau). \quad (\text{AU.19})$$

Every positive choice of τ gives a proved bound, and the infimum over such choices remains a bound. Multiplying all τ_j by the same positive number cancels between the two written determinant factors; it changes neither measure. No optimizer or numerical enclosure is assumed in (AU.19).

The lower determinant has the exact positive expansion

$$\det \mathbf{A}_q(\tau) = \sum_{\substack{I \subset \{0, \dots, q\} \\ |I|=q}} |\det(\mathbf{E}_q)_{[:,I]}|^2 \prod_{j \in I} \frac{\tau_j b^{2j}}{(2j)!}. \quad (\text{AU.20})$$

Expand each determinant multilinearly in columns in the product $\mathbf{E}_q D \mathbf{E}_q^*$: terms with repeated column indices vanish by alternation, and grouping permutations of the same subset gives (AU.20). Its subset $I = \{0, \dots, q-1\}$ is strictly positive by (AU.7). This expression retains full raw-jet minors and factorials without estimating an inverse by its individual entries.

4 The interval kernel and the actual comparison map

For $T > 0$, set

$$a_k(T) = a_k e^{-\alpha T} (d_k + T)^{-B}. \quad (\text{AU.21})$$

Restriction to $[-T, T]$ followed by the identity on polynomials maps the actual source to $L^2([-T, T], a_k(T) du)$ contractively. The same is true from $L^2(\nu_k)$, by the point-wise lower bound on that interval. Let $H_N^I(T)$ be the Lebesgue interval Gram. Then $a_k(T) H_N^I(T) \preceq H_N^\nu \preceq H_N$. Let P_j denote the Legendre polynomial specified by Rodrigues' formula, $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Integration by parts j times proves its orthogonality to lower degrees; direct evaluation of the resulting beta integral gives its squared norm $2/(2j+1)$. Its highest coefficient is $(2j)!/(2^j (j!)^2)$. The endpoint boundary terms vanish because $(x^2 - 1)^j$ has zeros of order j at both endpoints. Applying these constants to $P_j(u/T)$ gives the full interval jet kernel

$$(\mathbf{K}_N^I(T))_{(a,b,d),(a',b',e)} = \sum_{j=0}^N \frac{2j+1}{2T^{1+d+e}} P_j^{(d)}(\zeta_{ab}/T) \overline{P_j^{(e)}(\zeta_{a'b'}/T)}. \quad (\text{AU.22})$$

This follows by writing the inverse Gram in the orthogonal Legendre basis and applying (AU.6); each derivative contributes its stated T^{-d} . In particular

$$\mathbf{K}_{2q-1} \preceq \mathbf{K}_{2q-1}^\nu \preceq a_k(T)^{-1} \mathbf{K}_{2q-1}^I(T). \quad (\text{AU.23})$$

Equations (AU.14) and (AU.19) remain valid if their $\log \det \mathbf{K}_{2q-1}^\nu$ is replaced by $\log \det \mathbf{K}_{2q-1}^I(T) - q \log a_k(T)$. The global envelope version is at least as strong as each such interval version by the proved positive-matrix inequality (AU.23).

The restriction map has the exact orthogonal decomposition $f \mapsto (f1_{[-T,T]}, f1_{\mathbb{R} \setminus [-T,T]})$ in the original weighted L^2 source; its two squared norms add to the original norm. The interval bound drops a nonnegative term in that equality. It does not assign the outside component to external absence.

5 The explicit scalar bound and its finite gap

The following explicit scalar estimate is the independent full calculation VC.1–43, composed here with the preceding original quotient maps. Put $R_0 = \sqrt{\delta^2 + \gamma^2}$ and define

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = kR_0/T \leq 1/2, \quad N = 2q - 1, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \quad A_N = \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \\ B_q(r) &= \left(\frac{1+r}{1-r} \right)^q, \quad C_{k,q} = \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{AU.24})$$

All values are positive. In particular $q \geq k$ and $D \geq 2R_0$ give the stated bound on r . These choices bound constants; they do not change the original density.

Theorem 5.1 (A full finite Hermitian upper bound). *In the transported ordered quotient coordinates, and by the explicit congruence below also in the original S coordinates,*

$$\boxed{K_q \preceq K_{2q-1} \preceq \mathcal{C}_{k,q} K_q, \quad 0 \leq C_k \leq q \log \mathcal{C}_{k,q}.} \quad (\text{AU.25})$$

Proof. Let $\mathbf{T}_j$ be the coefficient matrix of the original map $\Psi(P) = P(c + iu)$ in degree j . Write $H_j^S, J_j^S, K_j^S, G_j^S$ for matrices in the original S monomials. Then

$$H_j^S = \mathbf{T}_j^* H_j \mathbf{T}_j, \quad J_j \mathbf{T}_j = \mathbf{T}_{q-1} J_j^S, \quad K_j = \mathbf{T}_{q-1} K_j^S \mathbf{T}_{q-1}^*.$$

The first identity is the source substitution, the second is monic division, and substitution in (AU.8) gives the third. Each $\det \mathbb{T}_j$ has modulus one, as proved in (AU.2). All source norms on S polynomials in this proof use H_j^S . The exact auxiliary coordinate map used in this proof is $\Psi_T P(x) = P(c + iTx)$, with inverse $f((S - c)/(iT))$. Its target measure is $Tm_k(Tx)dx$ and retains mass μ_h^k by substitution. The target monic relation is $\chi_T(x) = (iT)^{-q}\chi(c + iTx)$, so $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$. Monic division therefore proves $\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_\chi P$. At every original centre λ , the full order- d raw jet is $(\Psi_T P)^{(d)}((\lambda - c)/(iT)) = (iT)^d P^{(d)}(\lambda)$. The coefficient matrix has diagonal $(iT)^j$ at degree j ; its quotient determinant is $(iT)^{q(q-1)/2}$. Consequently the target quotient form is the original form transported by inverse congruence, and its determinant is $T^{-q(q-1)}V_j$ at every source degree j . This factor cancels in the central ratio, with all phases and masses still present in the map.

For a polynomial $f(x) = \sum f_j x^j$ put $\|f\|_{\text{coeff},1} = \sum |f_j|$. The Rodrigues formula above, expanded by the product rule, gives

$$P_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a.$$

The coefficient norm of a product is at most the product of the two coefficient norms by convolution and the triangle inequality. Since $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$, this proves $\|P_j\|_{\text{coeff},1} \leq \sum_a \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j$. The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$. In the orthogonal expansion $f = \sum_{j=0}^N f_j^L P_j$, Cauchy-Schwarz and the squared norm $2/(2j+1)$ give $|f_j^L| \leq \sqrt{(2j+1)/2} \|f\|_{L^2([-1,1])}$. Summing 4^j therefore proves

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}, \quad \deg f \leq N. \quad (\text{AU.26})$$

List all roots $z_1, \dots, z_q$ of χ_T , with repetitions; each has modulus at most r . Define e_a, h_j by

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j.$$

The second identity is formal multiplication of geometric series. It gives $|e_a| \leq \binom{q}{a} r^a$ and $|h_j| \leq \binom{q+j-1}{j} r^j$: the latter counts exponent lists of total degree j by their $q-1$ separating bars. For $0 \leq s \leq q-1$, multiply those inverse series through order s . It proves that the quotient of x^{q+s} by χ_T is exactly $Q_s = \sum_{j=0}^s h_j x^{s-j}$ and its remainder is $x^{q+s} - \chi_T Q_s$, of degree below q . Removing the leading term of $\chi_T Q_s$ removes precisely one from its coefficient norm. All its remaining coefficients are the negative remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \\ &\leq ((1+r)/(1-r))^q = B_q(r). \end{aligned}$$

At degrees below q the remainder is the original monomial and has coefficient norm one, also bounded by $B_q(r)$. Linearity proves

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{AU.27})$$

Every root was retained in this proof. Its smaller literal finite constant $\max\{1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1\}$ can also replace $B_q(r)$.

For v of degree at most $q-1$, each monomial is bounded by $\max\{1, |x|^{q-1}\}$. Using $\max(1, y) \leq 1 + y$ for $y \geq 0$ and the proved Laplace moment bound gives

$$\int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2. \quad (\text{AU.28})$$

On the other hand, restriction to $[-T, T]$ and (AU.21) prove $\|P\|_{H_N^S}^2 \geq Ta_k(T)\|\Psi_T P\|_{L^2([-1,1])}^2$. Take $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (AU.26)–(AU.28) and the exact division map now give

$$\|\text{rem}_{\chi} P\|_{H_{q-1}^S}^2 \leq C_{k,q} \|P\|_{H_N^S}^2. \quad (\text{AU.29})$$

The map estimated here is $s\pi_{\chi,N} : \mathcal{P}_N^S \rightarrow \mathcal{P}_{q-1}^S$, where $\pi_{\chi,N}$ is the polynomial quotient map and $s : \mathbb{C}[S]/(\chi) \rightarrow \mathcal{P}_{q-1}^S$ selects the unique remainder representative. Their exact identities are $\pi_{\chi,q-1}s = I$ and $\pi_{\chi,q-1}s\pi_{\chi,N} = \pi_{\chi,N}$. The composite $s\pi_{\chi,N}$ equals the identity on polynomials of degree below q .

Apply (AU.29) to the least lift $R_N^S x$ from the original matrices. Its remainder lifts the same x already at degree $q-1$, hence $x^* G_q^S x \leq x^* G_{q-1}^S x \leq C_{k,q} x^* G_N^S x$. Conversely degree inclusion gives $G_N^S \preceq G_q^S$. Conjugating by $(G_q^S)^{-1/2}$, bounding its positive eigenvalues and then inverting proves (AU.25), including the determinant exponent q , first for K_j^S and then by the displayed congruence for K_j . It also proves $C_{k,q} \geq 1$. $\square$

Retain the fixed constants

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_{\text{env}} = \max\{1, 2/(3c_h D)\}.$$

Because $bT \geq 2q$, the factorial ratio in (AU.24) is at most one. Also $M_k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that $U_k(T) \leq 3X^k$. The displayed A_N satisfies $A_{2q-1}^2 \leq (2q/9)256^q$. On $[0, 1/2]$ the derivative of $\log((1+r)/(1-r))$ is $2/(1-r^2) \leq 8/3$, so

$$\log B_q(r)^2 \leq \frac{16}{3}qr = \frac{16R_0}{3D}k. \quad (\text{AU.30})$$

Use also $\vartheta_h^{-(k-3)} \leq K^k$, $1 + Dq + k - 3 \leq 1 + (D+1)q$, and $e^{-3\alpha} \leq 1$. Substitution in (AU.24)–(AU.25) proves the finite scalar bound

$$\begin{aligned} 0 \leq C_k \leq \mathcal{U}_k &:= (\alpha D + \log 256)q^2 \\ &+ \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &+ Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \end{aligned}$$

(AU.31)

All of its constants are independent of k and q for fixed h, b . The exact finite bounds (AU.14), (AU.19), and $q \log C_{k,q}$ can be retained together by taking their minimum; each independently bounds the same C_k with the proved maps.

The complete central lower estimate in the balanced-window/factor-four source is $C_k \geq \mathcal{L}_k = 2(q-1)(\log k + \log(\delta/(2C_h^{\text{bal}})))$. The current explicitly computed gap is the literal function

$$\begin{aligned} \mathcal{U}_k - \mathcal{L}_k &= (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &+ Bq \log(1 + (D+1)q) + q \log C_{\text{env}} \\ &- 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \end{aligned} \quad (\text{AU.32})$$

This quantity concerns the two displayed upper and lower estimates. It is not an asserted exact value of the unknown C_k . Since $q = \ell(k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Dividing every term of (AU.31)–(AU.32) by q^2 proves

$$\lim_{k \rightarrow \infty} \frac{\mathcal{U}_k - \mathcal{L}_k}{q_k^2} = \alpha D + \log 256 > 0, \quad \limsup_{k \rightarrow \infty} C_k/q_k^2 \leq \alpha D + \log 256. \quad (\text{AU.33})$$

Dividing the upper estimate by $q \log k$ gives the exact leading factor $(\alpha D + \log 256)q / \log k$. It grows as $(\alpha D + \log 256)k^2 / \log k$ for $m = 1$ and as $(\alpha D + \log 256)(m - 1)k^3 / \log k$ for $m > 1$. The division step itself has only the linear-in- k logarithmic cost (AU.30). The two quadratic contributions in this scalar proof come from the full coefficient estimate (AU.26) and the least density over the interval in (AU.21). The positive matrices in (AU.14), (AU.19), and (AU.22) retain the directional information discarded in those scalar estimates, with their exact source and quotient maps already proved. No upper asymptotic for those sharper determinants is inserted.

6 A full-jet quadratic benchmark and its exact relation family

Here the auxiliary source is explicitly Lebesgue measure on $[-1, 1]$ and the auxiliary relation is u^q . Its mass is two. These data are inserted into the same sequence (AU.5), kernel (AU.8), and central ratio (AU.9). No assertion that they are an arithmetic density is made. Write D_d for its d by d monomial determinant, $D_0 = 1$, and $B_j(q)$ for the j by j determinant of the relation measure $u^{2q} du$ on that interval. The quotient-first triangular determinant calculation proving (AU.15) gives exactly

$$\frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} = \frac{D_{q+1} B_q(q)}{B_1(q) D_{2q}}, \quad B_1(q) = \frac{2}{2q+1}. \quad (\text{AU.40})$$

The orthogonal Legendre calculation gives

$$D_d = \prod_{j=0}^{d-1} \frac{2}{2j+1} \left(\frac{2^j}{\binom{2j}{j}} \right)^2. \quad (\text{AU.41})$$

For $q = 2s$, parity splits the relation Gram into two s by s blocks, and

$$B_{2s}(2s) = C_s(2s+1/2)C_s(2s+3/2), \quad C_s(x) = \frac{\prod_{j=0}^{s-1} (j!)^2}{\prod_{a,b=0}^{s-1} (x+a+b)}. \quad (\text{AU.42})$$

Indeed the even block has entries $1/(2s+a+b+1/2)$ and the odd block entries $1/(2s+a+b+3/2)$. The Cauchy determinant formula used here has numerator $\prod_{a<b} (b-a)^2 = \prod_j (j!)^2$. It follows by multiplying the determinant by all its denominators: the numerator is alternating separately in its two lists of variables, so contains their two Vandermonde factors. Its degree is their total degree, and comparing one residue inductively gives constant one. Substitution gives (AU.42) with no absolute-value or sign ambiguity.

Theorem 6.1 (An explicit quadratic scale in the full-origin jet fibre). *For every even integer $q \geq 2$,*

$$\left| \log \frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} - \kappa_* q^2 \right| \leq 12q \log(8q), \quad \kappa_* := \frac{9}{2} \log 3 - 6 \log 2 = \frac{3}{2} \log(27/16) > 0. \quad (\text{AU.43})$$

Proof. The elementary binomial bounds $4^j / (2j+1) \leq \binom{2j}{j} \leq 4^j$ hold because the central coefficient is the largest of the $2j+1$ coefficients summing to 4^j . Therefore (AU.41) gives

$$\log D_d = -d(d-1) \log 2 + E_d, \quad |E_d| \leq d \log(4d) \quad (d \geq 1). \quad (\text{AU.44})$$

At $s \geq 1$, expand the numerator of $C_s(2s)$ as $2 \sum_{i=1}^{s-1} (s-i) \log i$. The function $f(x) = (1-x) \log x$ is increasing on $(0, 1]$, has integral $-3/4$, and is nonpositive. Its right Riemann sum differs from its integral by a number between zero and $(\log s + 1)/s$: the difference is bounded above by $-\int_0^{1/s} f(x) dx \leq (\log s + 1)/s$, using $-f(x) \leq -\log x$. It follows that this numerator is $s^2 \log s - 3s^2/2$ plus an error of absolute value at most $s \log s + 2s$.

The denominator's logarithm is $s^2 \log s + \sum_{a,b=0}^{s-1} \log(2 + a/s + b/s)$. Both partial derivatives of the latter function lie in $[0, 1/2]$. Its left-square Riemann sum differs from its integral by at most $1/s$, hence the sum by at most s . Two integrations give

$$\int_0^1 \int_0^1 \log(2 + x + y) dx dy = 18 \log 2 - 9 \log 3 - 3/2.$$

For example use the second antiderivative $F(t) = t^2 \log t/2 - 3t^2/4$ and calculate $F(4) - 2F(3) + F(2)$. Thus

$$|\log \mathcal{C}_s(2s) - 9s^2 \log(3/4)| \leq s \log s + 3s.$$

For $c > 0$, the change to $\mathcal{C}_s(2s + c)$ has logarithm $-\sum_{a,b} \log(1 + c/(2s + a + b))$, between $-cs/2$ and zero. Using $c = 1/2, 3/2$ in (AU.42) proves

$$\left| \log B_q(q) - \frac{9}{2} q^2 \log(3/4) \right| \leq q \log(q/2) + \frac{7}{2} q. \quad (\text{AU.45})$$

Insert (AU.44)–(AU.45) into (AU.40). The quadratic coefficient is $3 \log 2 + (9/2) \log(3/4) = \kappa_*$. The remaining absolute error is at most

$$3q \log 2 + (q + 1) \log(4(q + 1)) + 2q \log(8q) \\ + q \log(q/2) + \frac{7}{2} q + \log((2q + 1)/2).$$

For $q \geq 2$, use $q + 1 \leq 3q/2$, $4(q + 1) \leq 8q$, $\log(q/2) \leq \log(8q)$ and $\log((2q + 1)/2) \leq \log(8q)$. The two constant multiples of q are bounded together by $3q \log(8q)$, since $3 \log(16) > 3 \log 2 + 7/2$. Their sum is then at most $8q \log(8q)$, hence certainly the displayed $12q \log(8q)$. This proves (AU.43) for every indicated finite q , not only along a limit. $\square$

There is an exact algebraic relation of this benchmark's quotient to every original grid quotient. Define

$$\psi_t(u) = \prod_{a,b} (u - t\zeta_{ab})^\ell, \quad \mathcal{A} = \mathbb{C}[t, u]/(\psi_t(u)). \quad (\text{AU.46})$$

Monic division over $\mathbb{C}[t]$ proves that $1, u, \dots, u^{q-1}$ form a free $\mathbb{C}[t]$ basis: division gives spanning and a nonzero multiple of a monic degree- q polynomial cannot have smaller degree. Every fibre has dimension q . The fibre at $t = 1$ is the original transported quotient; at $t = 0$ it is $\mathbb{C}[u]/u^q$. For $t \neq 0$,

$$\mathcal{A}_t \xrightarrow{\sim} \mathcal{A}_1, \quad [f(u)] \mapsto [f(tu)], \quad [g(u)] \mapsto [g(u/t)] \quad (\text{AU.47})$$

are inverse maps, since $\psi_t(tu) = t^q \psi_1(u)$. Their full raw jet equation is $(d/du)^d f(tu)|_{u=\zeta_{ab}} = t^d f^{(d)}(t\zeta_{ab})$, so the coefficient d in multiplication's raw-jet Jordan blocks and each order ℓ remain for $t \neq 0$. At $t = 0$ monic division gives the complete merged order q , rather than an assertion that separate evaluation rows stay invertible at coincident points.

For any fixed positive polynomial source H_N , remainder matrices $J_N(t)$ are polynomial in t by the monic subtraction algorithm. They contain I_q in their first q columns. Thus $K_N(t) = J_N(t)H_N^{-1}J_N(t)^*$ is positive for every real t , including zero, and the exact central function obeys

$$C(t) = \log \det K_{2q-1}(t) - \log \det K_q(t), \quad C'(t) = \text{Tr}(K_{2q-1}^{-1}K'_{2q-1} - K_q^{-1}K'_q), \quad (\text{AU.48})$$

where $K'_N = J'_N H_N^{-1} J_N^* + J_N H_N^{-1} (J'_N)^*$. The determinant derivative follows by its multilinear expansion. This gives both the free family and the exact finite source-dependent map along it. The estimate (AU.43) belongs to its stated Lebesgue special fibre. Equations (AU.46)–(AU.48) do not transfer that estimate to $t = 1$ without the displayed change in the full kernel.

7 Source representatives and support

The source polynomial maps throughout are the original $\mathcal{V}_{h,k}P = P(D_1 + \dots + D_k)F_h^{\otimes k}$, $D = -x\partial_x$, $\mathcal{M}F_h = g/h$. The arithmetic inclusion is the original $\eta[P] = [g/h]_h^{\otimes k}P(\sum s_i)$, after the explicit chart (AU.2). Each local Taylor constant of g/h at a selected complete zero is nonzero. Recursively solving multiplication of truncated Taylor series therefore proves this factor is a unit. It does not alter the original cyclic kernel (AU.4). The comparison to ν_k changes the norm used in an estimate, while its identity polynomial map leaves η , every raw jet, and the original relations unchanged.

For two polynomial representatives with the same quotient, monic division gives their difference χQ . In the tensor variables the original annihilator identity supplies $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$, with Q_i obtained by successive fixed-order monic divisions. The final remainder is zero because the monomial tensor basis is independent and the left side annihilates the sum. If $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ and $h(D)F_h = \Theta\phi_0$, the actual primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) Q(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{AU.49})$$

The tensor differential at position i contributes the second $(-1)^{i-1}$, cancelling the displayed one and giving exactly $\mathcal{V}_{h,k}(\chi Q)$. Thus every quotient comparison retains the same cohomology class with an explicit original primitive. On the supported carrier a fibre map sends (λ, v) to (λ, Lv) and sends external τ to τ . The killed relation has image $(\lambda, 0) = e_\lambda$. No comparison identifies that supported zero with external absence.

8 Verification and provenance

The complete arithmetic source (including the proof of its envelope), the exact endpoint-window source, the full PR24 note and scope correction/review, and the owner's complete balanced-window/factor-four proof were read for this calculation. Its new work is (AU.11)–(AU.23), the scalar estimate (AU.24)–(AU.33), and the full calculation (AU.40)–(AU.48). The constants $c_h, \vartheta_h, M_h(\pm b)$ retain their supplied definitions; no numerical enclosure of actual quartet data is claimed. The accompanying finite rational checker verifies the displayed maps and determinant formulas on declared auxiliary positive sources. It does not certify the analytic source by sampled values and makes no Lean claim.